BRIGHT TUITION CLASSES

Time : 3 hrs

Date:4/1/26

Time: 11:30 AM to 2:30 PM

NEET

M.M. : 720

STD : 12

Important Instructions :

1. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology**
   50 questions in each subject are divided into **two sections (A and B)** as per details given below:
   (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos.-1 to 35, 51 to 85, 101 to 135 and 151 to 185). All Questions are compulsory.
   (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos. – 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
   Candidates are advised to read all 15 questions in each subject of Section-B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
2. Each question carries **4 marks**. For each correct response, the candidate will get **4 marks**. For each incorrect response, **1 mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue / Black Ball point Pen only** for writing particulars on this page / marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must handover the Answer Sheet (ORIGINAL and OFFICE Copy)** to the **Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
7. Each candidate must show on-demand his/her Admit Card to the Invigilator.
8. No candidate, without special permission of the Centre Superintendent or Invigilator, would leave his/her seat.
9. Use of Electronic/Manual Calculator is prohibited.
10. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
11. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
12. The candidates will write the Correct Test Booklet Code as given in the Test Booklet / Answer Sheet in the Attendance Sheet.

Name of the Candidate (In CAPITALS) : .....

Roll Number : .....

OMR Bar Code Number : .....

Candidate's Signature : ..... Invigilator's Signature .....

1st FLOOR AMIN BOOK STORE ,NAVJEEVAN , AJWA ROAD VADODARA

(M)- 9824038852,9904751283

Bright Tuition Class

Subject : Physics
Standard : 11,12
Total Mark : 180

NEET full course 6 physics

Paper Set : 1
Date : 04-01-2026
Time : 3H:0M

Physics - Section A (MCQ)

(1) A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is $20\ m/s^2$ at a distance of $5\ m$ from the mean position. The time period of oscillation is

(A) $2\pi\ s$ (B) $\pi\ s$
(C) $1\ s$ (D) $2\ s$

(2) If the mass of the Sun were ten times smaller and the universal gravitational constant were ten times larger in magnitude, which of the following is not correct?

(A) Raindrops will fall faster.
(B) Walking on the ground would become more difficult.
(C) $'g'$ on the Earth will not change.
(D) Time period of a simple pendulum on the Earth would decrease.

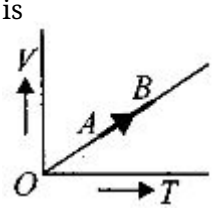
(3) Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area $3A$. If the length of the first wire is increased by Δl on applying a force F, how much force is needed to stretch the second wire by the same amount?

(A) $9F$ (B) $6F$
(C) F (D) $4F$

(4) A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of $27\,^oC$ two successive resonances are produced at $20\,cm$ and $73\,cm$ of column length. If the frequency of the tuning fork is $320\,Hz$, the velocity of sound in air at $27\,^oC$ is .... m/s

(A) 330 (B) 339
(C) 300 (D) 350

(5) The volume (V) of a monatomic gas varies with its temperature (T) , as shown in the graph. The ratio of work done by the gas , to the heat absorbed by it, when it undergoes a change from state A to state B , is



(A) $\frac{2}{5}$ (B) $\frac{2}{3}$
(C) $\frac{2}{7}$ (D) $\frac{1}{3}$

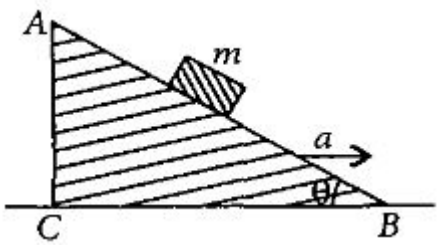
(6) At what temperature will the rms speed of oxygen molecules become just sufficient for escaping from the Earth's atmosphere ? (Given: Mass of oxygen molecule $(m) = 2.76 \times 10^{-26}\,kg$ Boltzmann's constant $k_B = 1.38 \times 10^{-23}\ JK^{-1}$)

(A) $2.508 \times 10^4\,K$ (B) $8.360 \times 10^4\,K$
(C) $1.254 \times 10^4\,K$ (D) $5.016 \times 10^4\,K$

(7) The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at $(2, 0, -3)$, about the point $(2, -2, -2)$, is given by

(A) $-8\hat{i} - 4\hat{j} - 7\hat{k}$ (B) $-4\hat{i} - \hat{j} - 8\hat{k}$
(C) $-7\hat{i} - 4\hat{j} - 8\hat{k}$ (D) $-7\hat{i} - 8\hat{j} - 4\hat{k}$

(8) A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration a towards the right. The relation between a and θ for the block to remain stationary on the wedge is

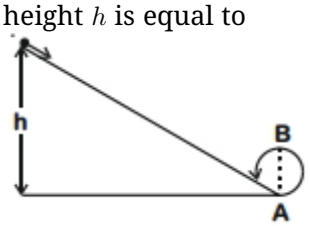


(A) $a = \frac{g}{cosec\theta}$ (B) $a = \frac{g}{sin\theta}$
(C) $a = gtan\ \theta$ (D) $a = gcos\ \theta$

(9) Which one of the following statements is incorrect?

(A) Rolling friction is smaller than sliding friction.
(B) Limiting value of static friction is directly proportional to normal reaction.
(C) Coefficient of sliding friction has dimensions of length.
(D) Frictional force opposes the relative motion.

(10) A body initially at rest and sliding along a frictionless track from a height h (as shown in the figure) just completes a vertical circle of diameter $AB = D$. The height h is equal to



(A) $\frac{3}{2}D$ (B) D
(C) $\frac{5}{4}D$ (D) $\frac{7}{5}D$

(11) A student measured the diameter of a small steel ball using a screw gauge of least count $0.001\,cm$. The main scale reading is $5\,mm$ and zero of circular scale division coincides with 25 divisions above the

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reference level. If screw gauge has a zero error of $-0.004\,cm$, the correct diameter of the ball is

(A) $0.521\,cm$ (B) $0.525\,cm$

(C) $0.529\,cm$ (D) $0.053\,cm$

(12) The power radiated by a black body is P and it radiates maximum energy at wavelength,λ_0. If the temperature of the black body is now changed so that it radiates maximum energy at wavelength $\frac{3}{4}\lambda_0$, the power radiated by it becomes nP. The value of n is

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$

(C) $\frac{81}{256}$ (D) $\frac{256}{81}$

(13) A thin diamagnetic rod is placed vertically between the poles of an electromagnet. When the current in the electromagnet is switched on, then the diamagnetic rod is pushed up, out of the horizontal magnetic field. Hence the rod gains gravitational potential energy. The work required to do this comes from

(A) The current source

(B) The magnetic field

(C) The induced electric field due to the changing magnetic field

(D) The lattice structure of the material of the rod

(14) The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom is

(A) $1:1$ (B) $1:-1$

(C) $1:-2$ (D) $2:-1$

(15) Current sensitivity of a moving coil galvanometer is $5\,div/mA$ and its voltage sensitivity (angular deflection per unit voltage applied) is $20\,div/V$. The resistance of the galvanometer is ..... Ω

(A) 40 (B) 25

(C) 500 (D) 250

(16) When the light of frequency $2v_0$ (where v_0 is threshold frequency), is incident on a metal plate , the maximum velocity of electrons emitted is v_1. When the frequency of the incident radiation is increased to $5v_0$, the maximum velocity of electrons emitted from the same plate is v_2. The ratio of v_1 to v_2 is

(A) $1:2$ (B) $1:4$

(C) $2:1$ (D) $4:1$

(17) An em wave is propagating in a medium with a velocity $\vec{v} = v\hat{i}$. The instantaneous oscillating electric field of this em wave is along $+y$ axis. Then the direction of oscillating magnetic field of the em wave will be along

(A) $-z$ direction (B) $+z$ direction

(C) $-x$ direction (D) $-y$ direction

(18) Unpolarised light is incident from air on a plane surface of a material of refractive index μ. At a particular angle of incidence i, it is found that the reflected and refracted rays are perpendicular to each other. Which of the following options is correct for this situation?

(A) Reflected light is polarised with its electric vector parallel to the plane of incidence

(B) Reflected light is polarised with its electric vector perpendicular to the plane of incidence

(C) $i = tan^{-1}\left(\frac{1}{\mu}\right)$

(D) $i = sin^{-1}\left(\frac{1}{\mu}\right)$

(19) The refractive index of the material of a prism is $\sqrt{2}$ and the angle of the prism is 30°. One of the two refracting surfaces of the prism is made a mirror inwards, by silver coating. A beam of monochromatic light entering the prism from the other face will retrace its path (after reflection from the silvered surface) if its angle of incidence on the prism is

(A) 60 (B) 45

(C) 0 (D) 30

(20) For a radioactive material, half-life is 10 minutes. If initially there are 600 number of nuclei, the time taken (in minutes) for the disintegration of 450 nuclei is

(A) 20 (B) 10

(C) 15 (D) 30

(21) In a $p-n$ junction diode, change in temperature due to heating

(A) affects only reverse resistance

(B) affects only forward resistance

(C) affects the overall $V-I$ characteristics of $p-n$ junction

(D) does not affect resistance of $p-n$ junction

(22) The magnetic potential energy stored in a certain inductor is $25\ mJ$, when the current in the inductor is $60\ mA$. This inductor is of inductance .....H

(A) 0.138 (B) 138.88

(C) 13.89 (D) 1.389

(23) A set of n equal resistors, of value R each, are connected in series to a battery of emf E and internal resistance R. The current drawn is I. Now, the n resistors are connected in parallel to the same battery. Then the current drawn from battery becomes $10\,I$. The value of n is

(A) 10 (B) 11

(C) 9 (D) 20

(24) An electron falls from rest through a vertical distance h in a uniform and vertically upward directed electric field E. The direction of electric field is now reversed, keeping its magnitude the same. A proton is allowed to fall from rest in it through the same vertical distance h. The time of fall of the electron, in comparison to the time of fall of the proton is

(A) smaller (B) 5 times greater

(C) equal (D) 10 times greater

(25) The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is

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(A) Independent of the distance between the plates

(B) Linearly proportional to the distance between the plates

(C) Inversely proportional to the distance between the plates

(D) Proportional to the square root of the distance between the plates

(26) The vector $\vec{P} = a\hat{i} + a\hat{j} + 3\hat{k}$ and $\vec{Q} = a\hat{i} - 2\hat{j} - \hat{k}$ are perpendicular to each other. The positive value of a is

(A) 3 (B) 4

(C) 9 (D) 13

(27) If the value of $'g'$ acceleration due to gravity, at earth surface is $10\,m/s^2$, its value in m/s^2 at the centre of the earth, which is assumed to be a sphere of radius 'R' metre and uniform mass density is

(A) 5 (B) $10/R$

(C) $10/2R$ (D) Zero

(28) A chimpanzee swinging on a swing in a sitting position, stands up suddenly, the time period will

(A) Become infinite (B) Remain same

(C) Increase (D) Decrease

(29) Velocity of a body on reaching the point from which it was projected upwards, is

(A) $v = 0$ (B) $v = 2u$

(C) $v = 0.5u$ (D) $v = u$

(30) The area of a rectangular field (in m^2) of length $55.3\,m$ and breadth $25\,m$ after rounding off the value for correct significant digits is

(A) 1382 (B) 1382.5

(C) 14×10^2 (D) 138×10^1

(31) $Assertion$: Perspiration from human body helps in cooling the body.
$Reason$: A thin layer of water on the skin enhances its emissivity.

(A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(C) If the Assertion is correct but Reason is incorrect.

(D) If both the Assertion and Reason are incorrect.

(32) In an orbital motion, the angular momentum vector is

(A) along the radius vector

(B) parallel to the linear momentum

(C) in the orbital plane

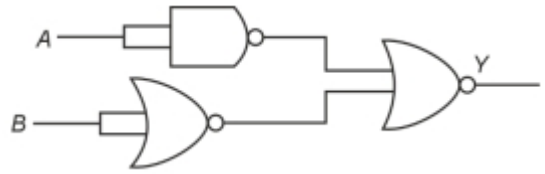
(D) perpendicular to the orbital plane

(33) A ball of mass $0.5\,kg$ is dropped from a height of $40\,m$. The ball hits the ground and rises to a height of $10\,m$. The impulse imparted to the ball during its collision with the ground is (Take g = 9.8 m/s^2)

(A) $21\,NS$ (B) $7\,NS$

(C) 0 (D) $84\,NS$

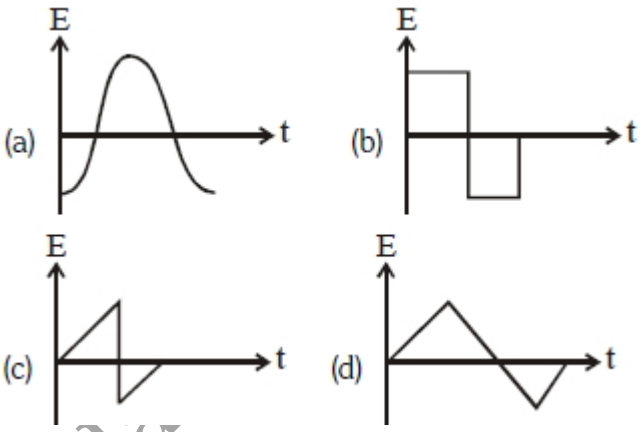
(34) The output (Y) of the given logic gate is similar to the output of an/a



(A) NOR gate (B) OR gate

(C) AND gate (D) $NAND$ gate

(35) The variation of EMF with time for four types of generators are shown in the figures. Which amongst them can be called AC ?



(A) (a) and (d) (B) $(a), (b), (c)$ and (d)

(C) (a) and (b) (D) Only (a)

(36) A transistor connected at common emitter mode contains load resistance of $5\,k\Omega$. If the input peak voltage is $5\,mV$ and the current gain is 50, find the voltage gain.

(A) 250 (B) 500

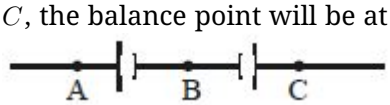
(C) 125 (D) 50

(37) The light rays having photons of energy $4.2\,eV$ are falling on a metal surface having a work function of $2.2\,eV$. The stopping potential of the surface is V

(A) 20 (B) 2

(C) 1.1 (D) 6.4

(38) A potentiometer is connected between A and B and the balance point is obtained at $203.6\,cm$. When the end of the potentiometer connected to B is shifted to C, then the balance point is obtained at $24.6\,cm$. If now the potentiometer be connected between B and C, the balance point will be at ..... cm



(A) 179 (B) 197.2

(C) 212 (D) 228

(39) An LED (Light Emitting Diode) is constructed from a $p-n$ junction based on a certain $Ga-As-P$ semi-conducting material whose energy gap is $1.9\,eV$. What is the wavelength of the emitted light?

(A) $650\,nm$ (B) $65\ \overset{o}{A}$

(C) $800\,nm$ (D) $8000\ \overset{o}{A}$

(40) $N-$ type semiconductors will be obtained, when germanium is doped with

(A) Phosphorus (B) Aluminium

(C) Arsenic (D) Both (a) or (c)

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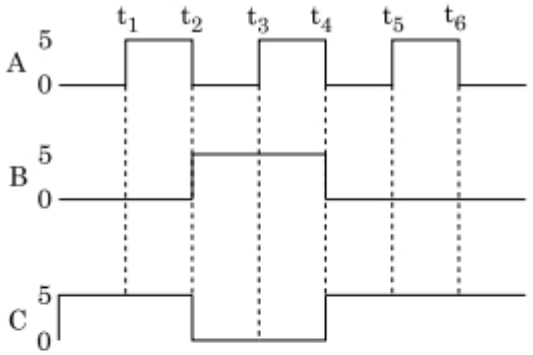
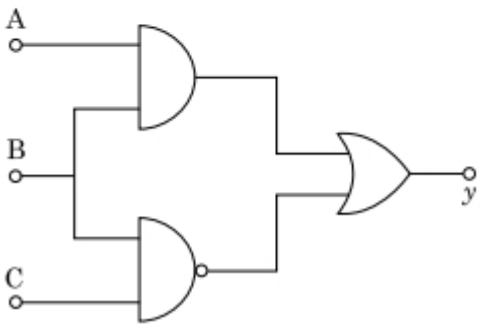
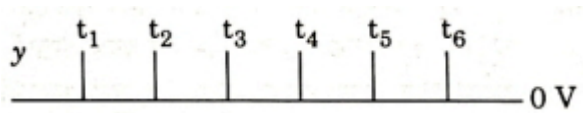
(41) For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y ?





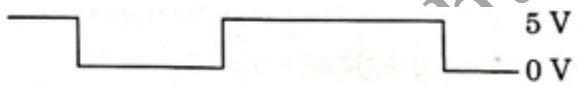
(A)



(B)

(C)

(D)

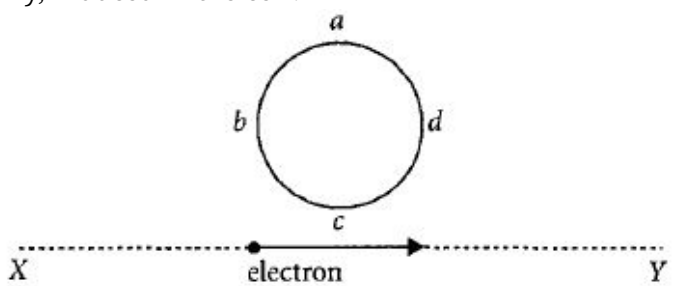


(42) In $N-$type semiconductors, majority charge carriers are

(A) Holes (B) Protons

(C) Neutrons (D) Electrons

(43) An electron moves on a straight line path XY as shown. The $abcd$ is a coil adjacent to the path of electron. What will be the direction of current, if any, induced in the coil?



(A) No current induced

(B) $abcd$

(C) $adcb$

(D) The current will reverse its direction as the electron goes past the coil.

(44) An electron collides elastically with $H-$like atom and excites it from ground state to $n = 3$. Find out the energy transfer to $H-$like atom in eV

(A) 12.1 (B) 10.2

(C) 12.75 (D) 10

(45) In case of linearly polarized light, the magnitude of the electric field vector:

(A) Does not change with time

(B) Varies periodically with time

(C) Increases and decreases linearly with time

(D) Is parallel to the direction of propagation

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Subject : Chemistry
Standard : 11,12
Total Mark : 180

12th Full Course NEET -6

Paper Set : 1
Date : 04-01-2026
Time : 0H:0M

Chemistry - Section A (MCQ)

(46) For electron in $'2s'$ and $'2p'$ orbitals, the orbital angular momentum values, respectively are :

(A) $\sqrt{2}\frac{h}{2\pi}$ and 0 (B) $\frac{h}{2\pi}$ and $\sqrt{2}\frac{h}{2\pi}$

(C) 0 and $\sqrt{6}\frac{h}{2\pi}$ (D) 0 and $\sqrt{2}\frac{h}{2\pi}$

(47) Match the LIST-I with LIST-II

	LIST-I		LIST-II
$A.$	Carbocation	$I.$	Species that can supply a pair of electrons.
$B.$	$C-$Free radical	$II.$	Species that can receive a pair of electrons.
$C.$	Nucleophile	$III.$	sp^2 hybridized carbon with empty p-orbital.
$D.$	Electrophile	$IV.$	sp^2/sp^3 hybridized carbon with one unpaired electron.

Choose the correct answer from the options given below :

(A) A-IV, B-II, C-III, D-I (B) A-II, B-III, C-I, D-IV

(C) A-III, B-IV, C-II, D-I (D) A-III, B-IV, C-I, D-II

(48) The alkane from below having two secondary hydrogens is:

(A) $4-$Ethyl$-3,4-$dimethyloctane

(B) $2,2,4,4-$Tetramethylhexane

(C) $2,2,3,3-$Tetramethylpentane

(D) $2,2,4,5-$Tetramethylheptane

(49) The equilibrium constant for decomposition of $H_2O(g)$
$H_2O(g) \rightleftharpoons H_2(g) + \frac{1}{2}O_2(g)$ $\left(\Delta G^\circ = 92.34\ kJmol^{-1}\right)$ is 8.0×10^{-3} at $2300\ K$ and total pressure at equilibrium is 1 bar. Under this condition, the degree of dissociation (α) of water is ..... $\times 10^{-2}$(nearest integer value).
[Assume α is negligible with respect to 1]

(A) 1 (B) 2

(C) 3 (D) 5

(50) Match the List $-I$ with List $-II$

	List $-I$ (Redox Reaction)		List $-I$ (Type of Redox Reaction)
(A)	$CH_{4(g)} + 2O_{2(g)} \xrightarrow{\Delta} CO_{2(g)} + 2H_2O_{(l)}$	(I)	Disproportionation reaction
(B)	$2NaH_{(s)} \xrightarrow{\Delta} 2Na_{(s)} + H_{2(g)}$	(II)	Combination reaction
(C)	$V_2O_{5(s)} + 5Ca_{(s)} \xrightarrow{\Delta} 2V_{(s)}5CaO_{(s)}$	(III)	Decomposition reaction
(D)	$2H_2O_{2(aq)} \xrightarrow{\Delta} 2O_{2(g)}O_{(l)}+$	(IV)	Displacement reaction

Choose the correct answer from the options given below :

(A) A-II, B-III, C-IV, D-I (B) A-II, B-III, C-I, D-IV

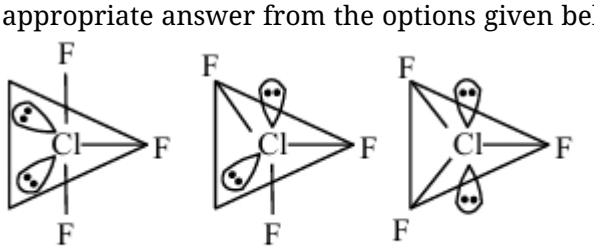
(C) A-III, B-IV, C-I, D-II (D) A-IV, B-I, C-II, D-III

(51) Identify correct statement/s :
(A) $-OCH_3$ and $-NHCOCH_3$ are activating group
(B) $-CN$ and $-OH$ are meta directing group
(C) $-CN$ and $-SO_3H$ are meta directing group
(D) Activating groups act as ortho $-$ and para directing groups
(E) Halides are activating groups
Choose the correct answer from the options given below :

(A) $(A),(C)$ and (D) only (B) $(A),(B)$ and (E) only

(C) (A) only (D) (A) and (C) only

(52) Given below are two statements:
Statement (I) : for $\ddot{C}\ell F_3$, all three possible structures may be drawn as follows. image
Statement (II) : Structure III is most stable, as the orbitals having the lone pairs are axial, where the ℓp $-$ bp repulsion is minimum.
In the light of the above statements, choose the most appropriate answer from the options given below:



I II III

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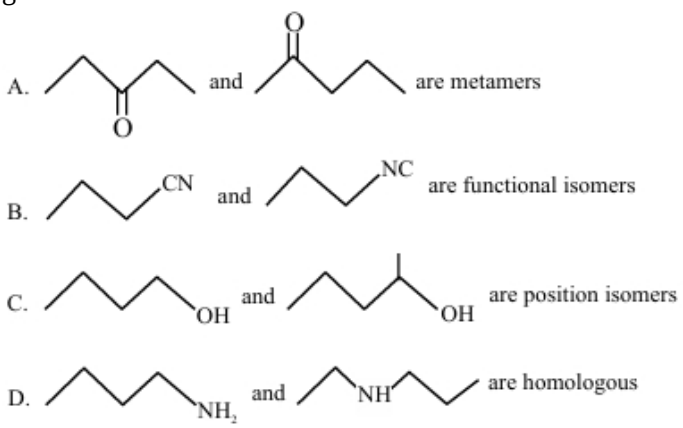
(A) Statement I is incorrect but statement II is correct.

(B) Statement I is correct but statement II is incorrect.

(C) Both Statement I and statement II are correct.

(D) Both Statement I and statement II are incorrect.

(53) Identify the correct statements from the following iamge Choose the correct answer from the options given below



(A) C & D only (B) B & C only

(C) A & B only (D) $A,\ B$ & C only

(54) Standard entropies of X_2, Y_2 and XY_5 are $70, 50$ and $110 JK^{-1}mol^{-1}$ respectively. The temperature in Kelvin at which the reaction $\frac{1}{2}X_2 + \frac{5}{2}Y_2 \rightarrow XY_5 \Delta H^- = -35 kJmol^{-1}$ Will be at equilibrium is _____ (Nearest integer)

(A) 600 (B) 700

(C) 800 (D) 300

(55) A weak acid HA has degree of dissociation x. Which option gives the correct expression of $pH = pK_a$?

(A) $\log(1+2x)$ (B) $\log\left(\frac{1-x}{x}\right)$

(C) 0 (D) $\log\left(\frac{x}{1-x}\right)$

(56) Consider the following reaction occurring in the blast furnace.
$Fe_3O_{4(s)} + 4CO_{(g)} \rightarrow 3Fe_{(l)} + 4CO_{2(g)}$ ' x ' kg of iron is produced when $2.32 \times 10^3 kg\ Fe_3O_4$ and $2.8 \times 10^2 kgCO$ are brought together in the furnace. The value of ' x ' is .....(nearest integer)
Given : Molar mass of $Fe_3O_4 = 232g\ mol^{-1}$
Molar mass of $CO = 28g\ mol^{-1}$
Molar mass of $Fe = 56g\ mol^{-1}$

(A) 420 (B) 320

(C) 220 (D) 250

(57) Choose the incorrect trend in the atomic radii (r) of the elements :

(A) $r_{Br} < r_K$ (B) $r_{Mg} < r_{Al}$

(C) r_{Rb} (D) $r_{At} < r_{Cs}$

(58) In Dumas' method for estimation of nitrogen, 0.5 gram of an organic compound gave $60\ mL$ of nitrogen collected at $300\ K$ temperature and $715\ mm$ Hg pressure. The percentage composition of nitrogen in the compound (Aqueous tension at 300 K $= 15$ mm Hg) is

(A) 1.257 (B) 20.87

(C) 18.67 (D) 12.57

(59) Given below are two statements :
Statement (I) : On nitration of m$-$xylene with HNO_3, H_2SO_4 followed by oxidation, $4-$nitrobenzene$-1,3-$dicarboxylic acid is obtained as the major product.
Statement (II) : CH_3 group is $o/p-$directing whileNO_2 group is m$-$directing group.
In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

(60) Given below are two statements :
Statement I : Dimethyl ether is completely soluble in water. However, diethyl ether is soluble in water to a very small extent.
Statement II : Sodium metal can be used to dry diethyl ether and not ethyl alcohol. In the light of given statements, choose the correct answer from the options given below

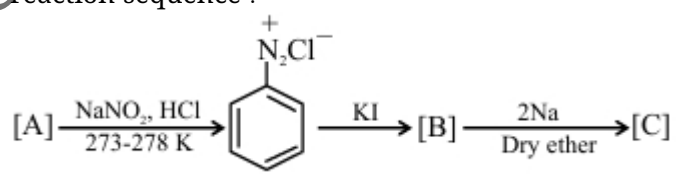
(A) Statement I is false but Statement II are true

(B) Both Statement I and Statement II are false

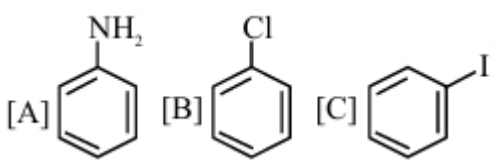
(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

(61) Identify [A], [B], and [C], respectively in the following reaction sequence :



(A)



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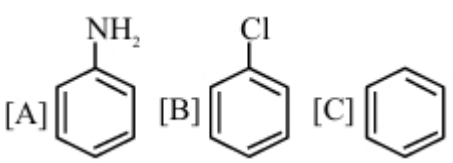
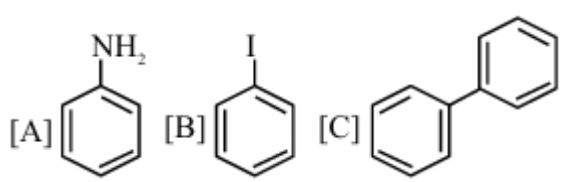
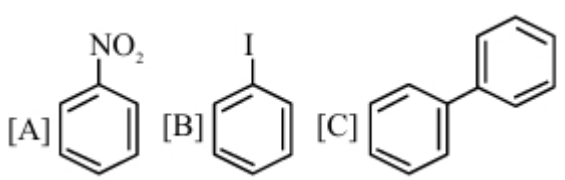


(C)



(D)



(62) Match List$-I$ with List$-II$:

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	List$-I$ Applications		List$-II$ Batteries/Cell
(A)	Transistors	(I)	Anode $-\ Zn/Hg$; Cathode $-\mathrm{HgO}+\mathrm{C}$
(B)	Hearing aids	(II)	Hydrogen fuel cell
(C)	Invertors	(III)	Anode $-Zn$; Cathode $-$ Carbon
(D)	Apollo space ship	(IV)	Anode $-Pb$; Cathode $-\mathrm{Pb}\mid \mathrm{PbO_2}$.

Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(B) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(D) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(63) Mixture of $1\ g$ each of chlorobenzene, aniline and benzoic acid is dissolved in $50\ mL$ ethyl acetate and placed in a separating funnel, $5\ M\ NaOH\ (30\ mL)$ was added in the same funnel. The funnel was shaken vigorously and then kept aside. The ethyl acetate layer in the funnel contains :

(A) benzoic acid

(B) benzoic acid and aniline

(C) benzoic acid and chlorobenzene

(D) chlorobenzene and aniline

(64) $\mathrm{HA(aq)} \rightleftharpoons \mathrm{H^+(aq)} + \mathrm{A^-(aq)}$
The freezing point depression of a $0.1\ m$ aqueous solution of a monobasic weak acid HA is $0.20°\mathrm{C}$. The dissociation constant for the acid is Given :
$\mathrm{K_f\,(H_2O)} = 1.8\ \mathrm{K\ kg\ mol^{-1}}$, molality $\equiv$ molarity

(A) 1.38×10^{-3} (B) 1.1×10^{-2}

(C) 1.90×10^{-3} (D) 1.89×10^{-1}

(65) Consider the following reactions. From these reactions which reaction will give carboxylic acid as a major product?

$(A)\ \mathrm{R-C\equiv N} \xrightarrow[\text{mild condition}]{\text{(i) } \mathrm{H^+/H_2O}}$

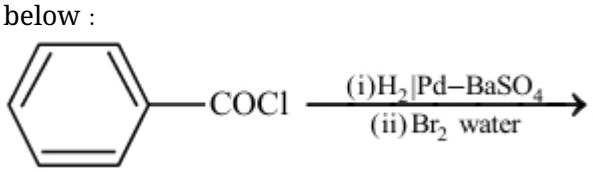
$(B)\ \mathrm{R-MgX} \xrightarrow[\text{(ii) } \mathrm{H_3O^+}]{\text{(i) } \mathrm{CO_2}}$

$(C)\ \mathrm{R-C\equiv N} \xrightarrow[\text{(ii) } \mathrm{H_3O^+}]{\text{(i) } \mathrm{SnCl_2/HCl}}$

$(D)\ \mathrm{R\cdot CH_2\cdot OH} \xrightarrow{\mathrm{PCC}}$

(E) : image

Choose the correct answer from the options given below :



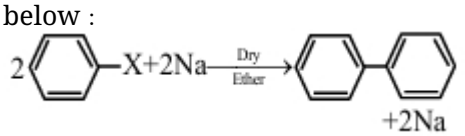
(A) A and D only (B) A, B and E only

(C) B, C and E only (D) B and E only

(66) Match List$-I$ with List$-II$

	List$-I$(Reaction)		List$-II$(Name of reaction)
(A)	image	(I)	Lucas reaction
(B)	$\mathrm{ArN_2^+X^-} \xrightarrow[\mathrm{HCl}]{\mathrm{Cu}} \mathrm{ArCl} + \mathrm{N_2}\uparrow + \mathrm{CuX}$	(II)	Finkelstein reaction
(C)	$\mathrm{C_2H_5Br} + \mathrm{NaI} \xrightarrow[\text{Acetone}]{\text{Dry}} \mathrm{C_2H_5I} + \mathrm{NaBr}$	(III)	Fittig reaction
(D)	$\mathrm{CH_3C(OH)\,(CH_3)\,CH_3} \xrightarrow[\mathrm{ZnCl_2}]{\mathrm{HCl}} \mathrm{CH_3C(Cl)\,(CH_3)\,CH_3}$	(IV)	Gatterman reaction

Choose the correct answer from the options given below :



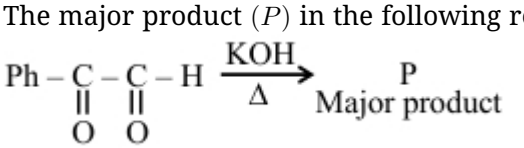
(A) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(B) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

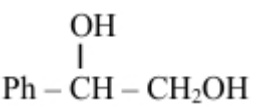
(C) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(D) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

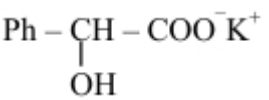
(67) The major product (P) in the following reaction is :



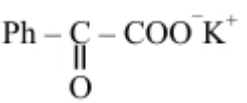
(A)



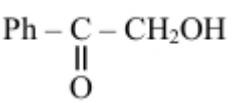
(B)



(C)



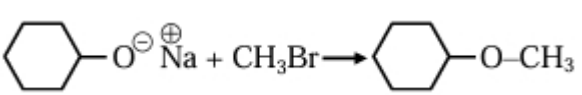
(D)



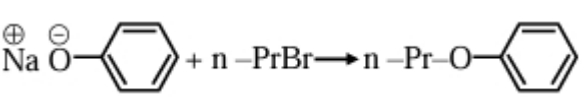
(68) Which one of the following reactions will not lead to the desired ether formation in major proportion? (iso$-$Bu $\Rightarrow$ isobutyl, sec$-$Bu $\Rightarrow$ sec$-$butyl, nPr $\Rightarrow$ n$-$propyl, ^{t}Bu $\Rightarrow$ tert$-$butyl, Et $\Rightarrow$ ethyl)

(A) $^{\mathrm{t}}\mathrm{Bu}\overset{\ominus}{\mathrm{O}}\overset{\oplus}{\mathrm{Na}} + \mathrm{EtBr} \rightarrow {}^{\mathrm{t}}\mathrm{Bu-O-Et}$

(B)



(C)



(D) iso$-$ $\mathrm{Bu}\overset{\ominus}{\mathrm{O}}\ \overset{\oplus}{\mathrm{Na}} + \mathrm{sec-BuBr} \rightarrow \mathrm{Sec-Bu-O-iso-Bu}$

(69) A group 15 element forms $d\pi - d\pi$ bond with transition metals. It also forms hydride, which is a strongest base among the hydrides of other group

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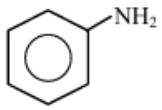
members that form $d\pi - d\pi$ bond. The atomic number of the element is .....

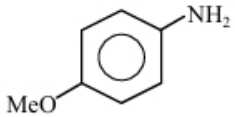
(A) 12 (B) 13

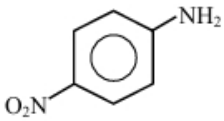
(C) 14 (D) 15

(70) The descending order of basicity of following amines is : image

Choose the correct answer from the options given below :

(A) 

(B) 

(C) 

(D) CH_3NH_2

(E) $(CH_3)_2NH$

(A) B > E > D > A > C (B) E > D > B > A > C

(C) E > D > A > B > C (D) E > A > D > C > B

(71) Given below are two statements

Statement I : The $N-N$ single bond is weaker and longer than that of P – P single bond

Statement II : Compounds of group 15 elements in $+3$ oxidation states readily undergo disproportionation reactions.

In the light of above statements, choose the correct answer from the options given below

(A) Statement I is true but statement II is false

(B) Both statement I and statement II are false

(C) Statement I is false but statement II is true

(D) Both statement I and statement II are true

(72) Which of the following ions is the strongest oxidizing agent ? [Atomic Number of $Ce = 58, Eu = 63, Tb = 65, Lu = 71$]

(A) Lu^{3+} (B) Eu^{2+}

(C) Tb^{4+} (D) Ce^{3+}

(73) An octahedral complex having molecular composition Co. $5NH_3.Cl^2.SO_4$ has two isomers A and B. The solution of A gives a white precipitate with $AgNO_3$ solution and the solution of B gives white precipitate with $BaCl_2$ solution. The type of isomerism exhibited by the complex is,

(A) Co–ordinate isomerism (B) Linkage isomerism

(C) Ionisation isomerism (D) Geometrical isomerism

(74) Identify correct conversion during acidic hydrolysis from the following :

(A) starch gives galactose.

(B) cane sugar gives equal amount of glucose and fructose.

(C) milk sugar gives glucose and galactose.

(D) amylopectin gives glucose and fructose.

(E) amylose gives only glucose.

Choose the correct answer from the options given below:

(A) (C), (D) and (E) only (B) (A), (B) and (C) only

(C) (B), (C) and (E) only (D) (B), (C) and (D) only

(75) The reaction $A_2 + B_2 \rightarrow 2$ AB follows the mechanism

$A_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A + A$ (fast) , $A + B_2 \xrightarrow{k_2} AB + B$ (slow) ,

$A + B \rightarrow AB$ (fast) The overall order of the reaction is :

(A) 1.5 (B) 3

(C) 2.5 (D) 2

(76) The first transition series metal $'M'$ has the highest enthalpy of atomisation in its series. One of its aquated ion (M^{nt}) exists in green colour. The nature of the oxide formed by the above M^{n-} ion is :

(A) neutral (B) acidic

(C) basic (D) amphoteric

(77) In a first order decomposition reaction, the time taken for the decomposition of reactant to one fourth and one eighth of its initial concentration are t_1 and $t_2(s)$, respectively. The ratio t_1/t_2 will :

(A) $\frac{4}{3}$ (B) $\frac{3}{2}$

(C) $\frac{3}{4}$ (D) $\frac{2}{3}$

(78) Match List $-I$ with List $-II$.

	List $-I$		List $-II$
(A)	Bronze	(I)	Cu, Ni
(B)	Brass	(II)	Fe, Cr, Ni, C
(C)	UK silver coin	(III)	Cu, Zn
(D)	Stainless Steel	(IV)	Cu, Sn

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(B) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(79) The nature of oxide (TeO_2) and hydride (TeH_2) formed by Te, respectively are :

(A) Oxidising and acidic (B) Reducing and basic

(C) Reducing and acidic (D) Oxidising and basic

(80) Based on the data given below:

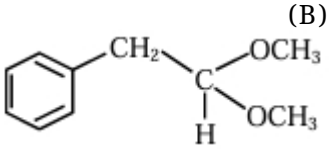
$E^0_{C_2O_7^{2-}/Cr^{3+}} = 1.33V$ $E^0_{Cl_2/Cl^{(-)}} = 1.36V$

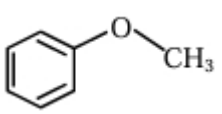
$E^0_{MnO_4^-/Mn^{2+}} = 1.51V$ $E^0_{Cr^{3+}/Cr} = -0.74V$ the strongest reducing agent is :

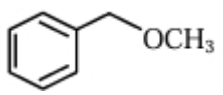
(A) Mn^{2+} (B) Cr

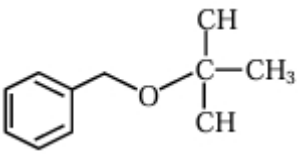
(C) MnO_4^- (D) Cl^-

(81) Which one of the following, with HBr will give a phenol?

(A) 

(B) 

(C) 

(D) 

(82) Which of the following acids is a vitamin?

(A) Adipic acid (B) Aspartic acid

(C) Ascorbic acid (D) Saccharic acid

(83) Given below are two statements :

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Statement I: In group 13, the stability of $+1$ oxidation state increases down the group.

Statement II: The atomic size of gallium is greater than that of aluminium.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

(84) The maximum number of orbitals which can be identified with $n = 4$ and $m_l = 0$ is

(A) 4 (B) 8

(C) 7 (D) 10

(85) The Molarity (M) of an aqueous solution containing 5.85 g of NaCl in 500 mL water is :

(Given : Molar Mass Na : 23 and Cl : $35.5 gmol^{-1}$)

(A) 20 (B) 0.2

(C) 2 (D) 4

(86) Given below are two statements :

Statement I : Gallium is used in the manufacturing of thermometers.

Statement II : A thermometer containing gallium is useful for measuring the freezing point (256 K) of brine solution.

In the light of the above statement, choose the correct answer from the options given below :

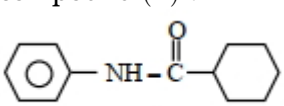
(A) Both Statement I and Statement II are false.

(B) Statement I is false but Statement II is true.

(C) Both Statement I and Statement II are true.

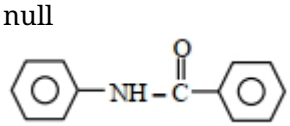
(D) Statement I is true but Statement II is false.

(87) Which of the following is metamer of the given compound (X) ?

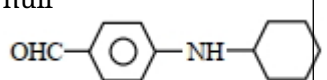


(X)

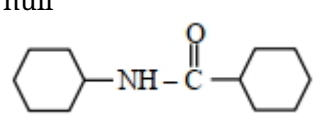
(A) null



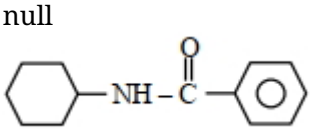
(B) null



(C) null



(D) null



(88) The correct statements from the following are :

(A) The decreasing order of atomic radii of group 13 elements is Tl $>$ In $>$ Ga $>$ Al $>$ B.

(B) Down the group 13 electronegativity decreases from top to bottom.

(C) Al dissolves in dil. HCl and liberate H_2 but conc. HNO_3 renders Al passive by forming a protective oxide layer on the surface.

(D) All elements of group 13 exhibits highly stable +1 oxidation state.

(E) Hybridisation of Al in $[Al(H_2O)_6]^{3+}$ ion is $sp^3 d^2$.

Choose the correct answer from the options given below:

(A) (C) and (E) only

(B) (A), (C) and (E) only

(C) (A), (B), (C) and (E) only

(D) (A) and (C) only

(89) At $-20°C$ and 1 atm pressure, a cylinder is filled with equal number of $H_2.I_2$ and HI molecules for the reaction

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, the K_p for the process is $x \times 10^{-1} \cdot x =$

[Given : $R = 0.082$ L atm K^{-1} mol^{-1}]

(A) 2 (B) 1

(C) 10 (D) 0.01

(90) The metal that shows highest and maximum number of oxidation state is:

(A) Fe (B) Mn

(C) Ti (D) Co

Bright Tuition Class

Bright Tution Class

Subject : Biology
Standard : 11,12
Total Mark : 180

Class 12 NEET Botany - 6

Paper Set : 1
Date : 04-01-2026
Time : 1H:0M

Biology - Section A (MCQ)

(91) Plant growth regulators, which are involved in growth inhibiting activities, also play an important role in –
(A) Pattern formation in plants
(B) Tropic growth in plants
(C) Cell division and cell enlargement
(D) Plant responses to wounds and stresses

(92) How many number of chromosomes are present in new formed child cells of human being at the end of mitosis?
(A) 23 (B) 46
(C) 92 (D) 184

(93) Which of the following induces femaleness in cucumber?
(A) Ethylene (B) Ethanol
(C) ABA (D) Gibberellin

(94) Match the following.

Column $-I$	Column $-II$
(1) Marginal	(p) Argemone
(2) Axile	(q) Dianthus
(3) Parietal	(r) Pea
(4) Free central	(s) Lemon

(A) $(1-p),(2-q),(3-r),(4-s)$
(B) $(1-q),(2-r),(3-s),(4-p)$
(C) $(1-r),(2-s),(3-p),(4-q)$
(D) $(1-s),(2-p),(3-q),(4-r)$

(95) First antibiotic isolated was
(A) Penicillin (B) Neomycin
(C) Terramycin (D) Streptomycin

(96) The best supporting fact that 'Viruses are living" is that viruses
(A) Are made of chemicals
(B) Cause diseases
(C) Duplicate themselves
(D) Penetrate cell membrane

(97) Photophosphorylation is process in which.
(A) CO_2 and O_2 bind with each other.
(B) Phosphoglyceric acid produced.
(C) Aspartic acid produced.
(D) Light energy converted into chemical energy and ATP are synthesized.

(98) Example of leaf for climbing.
(A) Onion (B) Grapes
(C) Cucumber (D) Pea

(99) When a cell is to divide, it synthesizes the various components including the genetic materials during interphase. Choose the correct option for synthesis of materials during interphase ?
(A) All types of DNAs of cell (nuclear and cytoplasmic) synthesis durig $S-$phase only
(B) Cytoplasmic growth occurs during G_{1-} phase only.
(C) Protein synthesis occurs during the all sub phases of interphase
(D) All the events of interphase are not genetically controlled

(100) Assertion : Both Endoplasmic reticulum and vacuoles are the part of endomembrane system.
Reason : Functions of both endoplasmic reticulum and vacuoles are not coordinated.
Options :
(A) Both Assertion and Reason are true and Reason is a correct explanation of the Assertion.
(B) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(C) Assertion is true but the Reason is False.
(D) Both Assertion and Reason are False.

(101) Which one of the following statements pertaining to plant structure is correct
(A) Cork have no stomata, but lenticels carry out transpiration
(B) Passage cells help in transfer of food from cortex to phloem
(C) Sieve tube elements posses cytoplasm but not nuclei
(D) The shoot apical meristem has a quiescent centre

(102) The incomplete breakdown of sugars in anaerobic respiration results in the formation of
(A) Fructose and water (B) Glucose and CO_2
(C) Alcohol and CO_2 (D) Water and C CO_2

(103) Intercellular space is more in..

(A) Collenchyma (B) Schlerenchyma

(C) Parenchyma (D) All correct

(104) One molecule of GTP is synthesised in the citric acid cycle, during conversion of :

(A) Citric acid to $\alpha-$ketoglutaric acid

(B) Succinic acid to succinyl CoA

(C) Succinyl CoA to succinic acid

(D) Malic acid to oxaloacetic acid

(105) Which of the following is a correct statement?

(A) Bacteria are exclusively heterotrophic organisms.

(B) Slime moulds are saprophytic organisms classified under Kingdom Monera.

(C) Mycoplasma have DNA, Ribosome and cell wall

(D) Cyanobacteria are a group of autotrophic organisms classified under Kingdom Monera.

(106) Characteristics which delimit a family are more general than those which delimit a

(A) Cohort (B) Phylum

(C) Class (D) Genus

(107) In dicot root

(A) Vascular bundles are scattered and with cambium

(B) Vascular bundles are arranged in a ring and have cambium

(C) Xylem and phloem radially arranged

(D) Xylem is always endarch

(108) Which one of the following is not the example of developmental heterophylly?

(A) Cotton (B) Buttercup

(C) Coriander (D) Larkspur

(109) Which of the following is responsible for mechanical support, enzyme transport and protein synthesis

(A) Dictyosomes (B) Cell membrane

(C) Mitochondria (D) Endoplasmic reticulum

(110) Which are green house crops ?

(A) Tomatoes and bellpepper

(B) Tomatoes and rice

(C) Banana and wheat

(D) Bell pepper and Neem

(111) Prop roots of Banyan tree are meant for

(A) Respiration

(B) Absorption of water from soil

(C) Retention of water in soil

(D) Providing support to big tree

(112) Which one of the following statements is wrong?

(A) Algae increase the level of dissolved oxygen in the immediate environment.

(B) Algin is obtained from red algae, and carrageenan from brown algae.

(C) Agaragar is obtained from Gelidium andGracilaria.

(D) Laminaria and Sargassum are used as food.

(113) If you are asked to classify the various algae into distinct groups, which of the following characters you should choose?

(A) Nature of stored materials in the cell

(B) Structural organization of thallus

(C) Chemical compolition of the cell wall

(D) Types of pigments present in the cell

(114) Find out correct pair.

(A) Klinefelter's syndrome - additional copy of X chromosome is absent.

(B) Turner's syndrome - one X -sex chromosome present

(C) Down's syndrome - monosomy of 21th pair chromosome

(D) Beta thalassemia - gene present on 5th pair of autosomal chromosomes

(115) Recessive traits are seen due to:

(A) Formation of non functional enzyme

(B) Enzyme is not produced

(C) Both (1) & (2)

(D) Formation of functional enzyme

(116) Match List $-I$ with List$-II$

	List$-I$		List$-II$
(A)	Two or more alternative forms of a gene	(I)	Back cross
(B)	Cross of F_1 progeny with homozygous recessive parent	(II)	Ploidy
(C)	Cross of F_1 progeny with any of the parents	(III)	Allele
(D)	Number of chromosome sets in plant	(IV)	Test cross

Choose the correct answer from the options given below:

(A) A - I , B - II , C - III , D - IV

(B) A - II , B - I , C - III , D - IV

(C) A - II , B - IV , C - I , D - II

(D) A - IV , B - III , C - II , D - I

(117) Cryopreservation of gametes of threatened species in viable and fertile condition can be referred to as

(A) in situ conservation by sacred groves

(B) in situ cryoconservation of biodiversity

(C) in situ conservation of biodiversity

(D) advanced ex situ conservation of biodiversity.

(118) Removal of RNA polymerase I from nucleoplasm will affect the synthesis of

(A) $r-RNA$ (B) $hnRNA$

(C) $t-RNA$ (D) $m-RNA$

(119) Number of sex chromosomes is normal in

(A) Super female (B) Turner's syndrome

(C) Klinefelter's syndrome (D) Down's syndrome

(120) India has only ..... percent of the word's land area, its share of the global species diversity is ..... percent.

(A) $8.1\%, 2.4\%$ (B) $7.6\%, 2.9\%$

(C) $2.4\%, 8.1\%$ (D) $5.9\%, 2.2\%$

(121) Choose incorrect sentence.

(A) A typical nucleosome contains 200 bp of DNA helix.

(B) Nucleosomes constitute the repeating unit of a structure in nucleus called chromatin thread like stained (coloured) bodies seen in nucleus

(C) In a typical nucleus, some region of chromatin are loosely packed and are referred to heterochromatin.

(D) Euchromatin is said to be transcriptonally active chromatin, whereas heterochromatinis inactive.

(122) RNA polymerase is involved in

(A) Translation (B) Transcription

(C) Translocation (D) Replication

(123) Match List$-I$ with List$-II$.

List$-I$	List$-II$
A. Alfred Hershey and Martha Chase	I. Streptococcus pneumoniae
B. Euchromatin	II. Densely packed and dark$-$stained
C. Frederick Griffith	III. Loosely packed and light$-$stained
D. Heterochromatin	IV. DNA as genetic material confirmation

Choose the correct answer from the options given below :

(A) A-II, B-IV, C-I, D-III (B) A-IV, B-II, C-I, D-III

(C) A-IV, B-III, C-I, D-II (D) A-III, B-II, C-IV, D-I

(124) It is the expression of y gene in lac-operon

(A) Beta galactosidase (B) Permease

(C) Transacetylase (D) Lactase

(125) After penetrating stigmatic and styler tissues, the pollen tube usually grow down towards the egg because

(A) The egg cell attracts the pollen tube as they have dissimilar electric charge

(B) The filiform apparatus of synergids is believed to attract the pollen tube

(C) It has no other passage to follow

(D) It grows under control of nucleus

(126) If diploid chromosome number in a flowering plant is 12, then which one of the following will have only 6 chromosomes

(A) Endosperm (B) Leaf cells

(C) Cotyledons (D) Synergids

(127) The organism which is useful to producechemical which inhibit immunity during organ transport.

(A) Streptococcus

(B) Monascus purpureus

(C) Lactobacillus

(D) Trichoderma polysporum

(128) Embryo sac is

(A) Megasporangium (B) Megaspore

(C) Female gametophyte (D) Female gamete

(129) Yeast have been used for the commercialproduction of

I. ethanol II. bread III. Incilage

Choose the correct option

(A) I and II (B) I and III

(C) I, II and III (D) None of these

(130) Assertion : In sigmoid growth curve, population finally stabilizes itself.

Reason : Finally, the death rate increases than the birth rate.

(A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(C) If the Assertion is correct but Reason is incorrect.

(D) If both the Assertion and Reason are incorrect.

(131) Choose the correct statement.

(A) Transcription and translation occur in same compartment for prokaryotes

(B) Monocistonic RNA - more than one structural genes under single promoter

(C) Introns and exons both code for protein synthesis

(D) In prokaryotes, splicing and tailing occurs before translation.

(132) Bird females have chromosome arrangement as

(A) XY (B) XO

(C) WZ (D) WW

(133) Which of the following is the most stable ecosystem

(A) Mountain (B) Desert

(C) Forest (D) Ocean

(134) In the logistic growth equation $dN/dt = rN\left[\frac{K-N}{K}\right]$, ' r ' represents : –

(A) Extrinsic rate of natural increase

(B) The base of natural logarithms

(C) Intrinsic rate of natural increase

(D) Carrying capacity

(135) Assertion : A network of food chains existing together in an ecosystem is known as food web.
Reason : An animal like kite cannot be a part of a food web.

(A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(C) If the Assertion is correct but Reason is incorrect.

(D) If both the Assertion and Reason are incorrect.

Bright Tution Class

Subject : Biology
Standard : 11,12
Total Mark : 180

Class 12 NEET Zoology - 6

Paper Set : 1
Date : 05-01-2026
Time : 1H:0M

Biology - Section A (MCQ)

(136) Vasopressin is related with
(A) Dilution of urine (B) Quick digestion
(C) Concentration of urine (D) Slow heart beat

(137) The last two pairs of ribs are named floating ribs because
(A) Their sternal parts are attached to the sternum directly
(B) Their sternal parts are attached on the 7th pair of ribs
(C) Their sternal parts remain free and do not even reach the sternum
(D) They float in the body cavity

(138) In DNA guanine pairs with
(A) Cytosine (B) Thymine
(C) Uracil (D) Adenine

(139) The non-protein enzyme is
(A) Ribozyme (B) Pepsin
(C) Trypsin (D) All of the above

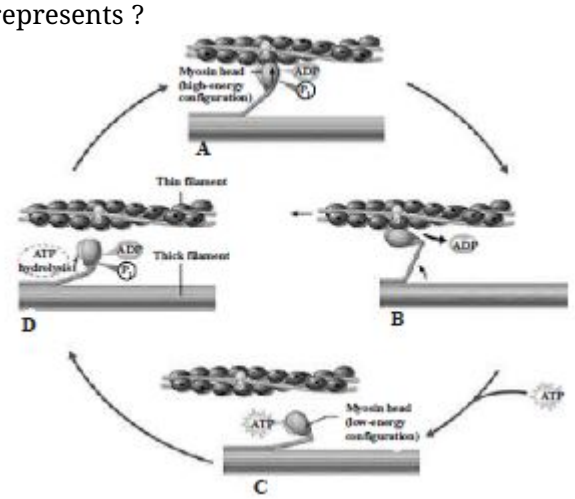
(140) Following are the statements with reference to 'lipids'.
(a) Lipids having only single bonds are called unsaturated fatty acids.
(b) Lecithin is a phospholipid.
(c) Trihydroxy propane is glycerol.
(d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
(e) Arachidonic acid has 16 carbon atoms.
Choose the correct answer from the options given below.
(A) (a) and (b) only (B) (c) and (d) only
(C) (b) and (c) only (D) (b) and (e) only

(141) Pneumotaxis centre is associated with
(A) Breathing (B) Respiration
(C) Movement (D) Closure of glottis

(142) Which of the following is not the function of conducting part of respiratory system ?
(A) Inhaled air is humidified
(B) Temperature of inhaled air is brought to body temperature
(C) Provides surface for diffusion of O_2 and CO_2
(D) It clears inhaled air from foreign particles

(143) All functions of the body are regulated and integrated by
(A) Respiratory system
(B) Digestive system
(C) Neuroendocrine system
(D) Excretory system

(144) The given figure represents the cross bridge cycle in skeletal muscle. What does the step B in the figure represents ?



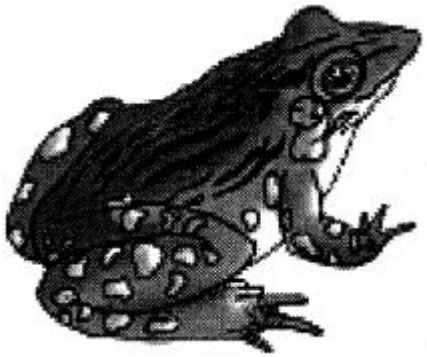
(A) Attachment of myosin head to actin forming cross bridge.
(B) Release of phosphate. Myosin changes shape to pull actin
(C) Attachment of new ATP to myosin head. The cross bridge detaches
(D) Splitting of ATP into ADP and Pi. Myosin cocks into its high energy conformation.

(145) Match List I with List II.

List I	List II
A. Mast cells	I. Ciliated epithelium
B. Inner surface of bronchiole	II. Areolar connective tissue
C. Blood	III. Cuboidal epithelium
D. Tubular parts of nephron	IV. Specialised connective tissue

Choose the correct answer from the options give below:
(A) A-III, B-IV, C-II, D-I (B) A-I, B-II, C-IV, D-III
(C) A-II, B-III, C-I, D-IV (D) A-II, B-I, C-IV, D-III

(146) Which is correct for given figure ?



(A) Notochord converted into spinal cord.

(B) Notochord not produced.

(C) Germinal layers do not made.

(D) Notochord only found during embryonicperiod.

(147) It is absent in sebum.

(A) Wax (B) Hydrocarbons

(C) Sterols (D) $NaCl$

(148) Dinucleotide is obtained by joining of two nucleotides together by phosphodiester linkage between which carbon of sugar of nucleotides are these linkages present?

(A) $5'$ and $3'$ (B) $5'$ and $5'$

(C) $3'$ and $3'$ (D) $1'$ and $5'$

(149) Diabetes is a disease in which the urine contains

(A) Sugar (B) Salt

(C) Fat (D) Protein

(150) Chief distinguishing features of the mammals are

(A) Hairy skin and ovipary

(B) Hairy skin and mammary glands

(C) Mammary glands and teeth

(D) Pinna and teeth

(151) Blood to the heart is brought by the venous system by

(A) Innominate and anterior abdominal

(B) Subclavian alone

(C) Vena cavae

(D) Anterior abdominal alone

(152) Radula is a file like rasping organ for feeding in $-$

(A) Echinodermata (B) Annelida

(C) Mollusca (D) Hemichordata

(153) True coelom appeared first in the course of evolution in

(A) Echinodermata (B) Annelida

(C) Chordata (D) Aschelminthes

(154) What is correct for association area of cerebral cortex ?

(A) Intersensory association, memory and communication.

(B) Memory, Emotional, reaction, Reflex action

(C) Biological clock, Sexual behavior, memory

(D) Communication, biological clock intersensory communication

(155) Rate of breating is controlled by

(A) The amount of freely available oxygen

(B) Carbon dioxide

(C) Muscular function of the body

(D) Stress

(156) ... exerts the greatest control over the body is responsiveness to changing conditions.

(A) Connective tissue (B) Neural tissue

(C) Bone (D) Cartilage

(157) Assertion: The thyroid gland is composed of follicles and stromal tissue.
Reason: Iodine is essential for the normal rate of hormone synthesis in the thyroid.

(A) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(B) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(C) Assertion is True but the Reason is False.

(D) Both Assertion & Reason are False.

(158) Match the following:

Column $-I$	Column $-II$	Column $-III$
X. RBC	P. Nucleated	A. $5-5.5$ Million
Y. Monocytes	Q. granulocyte	B. $2-3\%$
Z. Eosinophil	R. Lack of nucleus	C. $6-8\%$
	S. Lack of Hb	D. $20-25\%$

(A) $X-R-A, Y-S-D, Z-Q-B$

(B) $X-Q-B, Y-R-A, Z-P-C$

(C) $X-P-A, Y-P-D, Z-Q-B$

(D) $X-R-A, Y-P-C, Z-Q-B$

(159) Lactational amenorrhea is effective only upto a maximum period of ..... following parturition.

(A) One month (B) Two month

(C) Six month (D) Three month

(160) Cultivation of Bt cotton has been much in the news. The prefix Bt means

(A) barium-treated cotton seeds

(B) bigger thread variety of cotton with better tensile strength

(C) produced by biotechnology using restriction enzymes and ligases

(D) carrying an endotoxin gene from Bacillus thuringiensis

(161) Upon exposure to UV radiation, DNA stained with ethidium bromide will show

(A) Bright orange colour (B) Bright red colour

(C) Bright blue colour (D) Bright yellow colour

(162) Pithecanthropus erectus fossil was found in

(A) China (B) Japan

(C) Java (D) Texas

(163) Main cause of population explosion in the world is

(A) Excellent job facilities

(B) Increase in agricultural production

(C) Excellent health care

(D) Fewer battles and wars

(164) Which of the following $STDs$ are not curable?

(A) Gonorrhoea, Trichomoniasis, Hepatitis B

(B) Genital herpes, Hepatitis B, HIV infection

(C) Chlamydiasis, Syphilis, Genital warts

(D) HIV, Gonorrhoea, Trichomoniasis

(165) Daily oral contraceptive pill is

(A) Mala C (B) Mala N and Mala D

(C) Mala A (D) Mala D

(166) Vaginal orifice, urethral orifice and clitoris are protected by

(A) Labia majora (B) Labia minora

(C) Vulva (D) Anus

(167) Which is correct for oogenesis

(A) Oogonia $\rightarrow$ secondary oocyte $\rightarrow$ ovum

(B) Oogonia $\rightarrow$ primary oocyte $\rightarrow$ Ovum

(C) Ovum $\rightarrow$ oogonia $\rightarrow$ secondary oocyte $\rightarrow$ primary oocyte

(D) Oogonia $\rightarrow$ primary oocyte $\rightarrow$ secondary oocyte $\rightarrow$ ovum

(168) Which of the following is not the application of PCR?

(A) Detection of very low concentration of bacteria or virus

(B) Detection of mutations in genes in suspected cancer patients

(C) Amplification of desired DNA segment

(D) Detection of antibodies synthesised against pathogens

(169) Match the column

Substrate	Enzyme
(A) Ribonucleotide	(i) Chitinase
(B) Chitin	(ii) Cellulase
(C) Cellulose	(iii) Ribonuclease

(A) $A-i, B-ii, C-iii$ (B) $A-iii, B-i, C-ii$

(C) $A-iii, B-ii, C-i$ (D) $A-ii, B-i, C-iii$

(170) 'Pathogens' are

(A) Substances produced against any disease causative

(B) Chemical substances produced by the host cells to kill the parasite animal

(C) Disease spreading factors

(D) Cells which kill the parasites

(171) Match List I with List II.

List I	List II
A. Heroin	I. Effect on cardio-vascular system
B. Marijuana	II. Slow down body function
C. Cocaine	III. Painkiller
D. Morphine	IV. Interfere with transport of dopamine

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II (B) A-II, B-I, C-IV, D-III

(C) A-I, B-II, C-III, D-IV (D) A-IV, B-III, C-II, D-I

(172) $ECORI$ derives from .....

(A) Escherichia coli RI (B) Escherichia coli $RI\,13$

(C) Escherichia coli $RX\,13$ (D) Escherichia coli $RY\,13$

(173) Early Greek thinkers thought units of likecalled ..... were transferred to different planets including earth.

(A) Nitrogen based (B) Spore

(C) Pigments (D) Branching descent

(174) 'Rosie' cow known to produce a type of milk which has all the following characteristics

I. Protein content of $2.4\ g/L$.

II. Human Alpha lactalbumin

III. More nutritionally balanced for human babies than natural cow milk

Which of the above statements are correct ?

(A) I and II only (B) I and III only

(C) II and III only (D) I, II and III

(175) Plant cells are bombarded with high velocity micro - particles of ..... coated with DNA in a method known as gene gun.

(A) Sliver or platinum (B) Platinum or zinc

(C) Silicon or platinum (D) Gold or tungsten

(176) Match the following columns and select the correct option.

Column$-I$	Column$-II$
(a) Placenta	(i) Androgens
(b) Zona pellucida	(ii) Human Chori-onic Gonadotropin (hCG)
(c) Bulbo$-$urethral glands	(iii) Layer of the ovum
(d) Leydig cells	(iv) Lubrication of the Penis

(a) (b) (c) (d)

(A) (ii) (iii) (iv) (i) (B) (iv) (iii) (i) (ii)

(C) (i) (iv) (ii) (iii) (D) (iii) (ii) (iv) (i)

(177) Match the following techniques or instruments with their usage:

Column - I	Column - II
(a) Bioreactor	(i) Separation of DNA fragments
(b) Electrophoresis	(ii) Production of large quantities of products
(c) PCR	(iii) Detection of pathogen, based on antigen-antibody reaction
(d) $ELISA$	(iv) Amplification of nucleic acids

Select the correct option from following:

(A) $(a)-(ii),(b)-(i),(c)-(iii),(d)-(iv)$

(B) $(a)-(iii),(b)-(ii),(c)-(iv),(d)-(i)$

(C) $(a)-(ii),(b)-(i),(c)-(iv),(d)-(iii)$

(D) $(a)-(iv),(b)-(iii),(c)-(ii),(d)-(i)$

(178) In a mixture, DNA fragments are separated by

(A) Polymerase chain reaction

(B) Bioprocess engineering

(C) Restriction digestion

(D) Electrophoresis

(179) They probably live in east Africa grass landtwo mya.

(A) Dryopithecus (B) Ramapithecus

(C) Homo erectus (D) Australopithecus

(180) Restriction endonuclease is used in

(A) Genetic engineering (B) Tissue culture

(C) Cell fractionation (D) Regeneration of tissues